Quantum Computing: The next frontier in 10 years

There are three big recent developments that are positioning quantum computing and informatics to the forefront of disruptive changes over the next decade. China has recently started investing another $10 billion in Heifei for the National Laboratory for Quantum Information Sciences. In December 2018, President Donald Trump signed a bipartisan bill in the United States and enacted the National Quantum Initiative (NQI). The bill serves to lay out the plan for the U.S. to advance quantum computing. Specifically, it suggests U.S. research funding institutions such as the National Science Foundation and Department of Energy to direct funding to the tune of $1 billion dollars in improving quantum computing capabilities. Besides support from the government, there is an extensive amount of private investment that are also entering into the space. For example, the founder of Blackberry has contributed about $100 million to the Institute for Quantum Computing in Canada. This means that because of the renewed attention to this domain, we should be expecting dramatic advances in these technologies.

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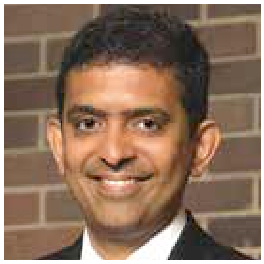
Second, we are at the cusp of many successes. The aforementioned investments are also indicative of the potential from the early successes we have seen in this field. The MIT Technology Review qubit (quantum equivalent of classical bits used for performing information processing) counter keeps track of current compute capacity of the latest quantum computers. Rigetti's quantum computers were able to do 128-qubit calculations as of August 2018. What started out as two-qubit compute in 1998 has grown into being 128-qubit compute in 2018, which is really quite remarkable.

Third, private companies are already building their capabilities to take advantage of the revolution. The obvious first line of companies will be the technology ones such as Microsoft, Google and Intel. Aerospace and defense companies such as Boeing and Lockheed Martin are among the ones that are seeing value from these fast computations. Of late, we are noticing even traditional mobility companies such as Ford and Volkswagen investing in quantum computing in conjunction with other partners. Quantum computing is useful for these automobile companies for activities such as autonomous vehicle routing and fleet management.

To see the value from quantum computing, consider autonomous vehicle routing. The vehicles should be able to consider combinations of routes and direct traffic accordingly. This leads to an explosion of options that a computer must consider for optimal solutions. Current classical computers are not efficient in these kinds of combinatorial calculations, and the quantum ones can be. For problems like this in logistics, product portfolios, etc., quantum computing can quickly provide optimal solutions.



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While the application domains relevant to quantum computing is enormous, quantum computing is not widely adopted yet. Why? One of the big open questions has always been that, while quantum computing is theoretically shown to perform better than classical computing, there has been little evidence to show this in action. This is primarily because engineering a quantum computer with the number of qubits required to prove their advantage over the most powerful classical supercomputers has proven to be an extremely challenging endeavor. Namely, the quantum hardware needs to be maintained in a fragile quantum state, which is not affected by environmental noise to function as a quantum computer. This, in turn, typically requires placing qubits in extreme conditions, such as in a millikelvin temperature environment (colder than outer space). However, with increased investments and interests from academia and industry, the number of qubits has seen a steady growth, with many believing that we are close to seeing the first examples of a quantum advantage. Also, crowdsourced competitions from companies such as Rigetti can further fuel this effort.

Anytime you have something good, you also have something bad. So, there may be some dramatic changes to the digital space because of quantum computing. Current implementations of cryptography are based on classical computers' inability to decrypt the encryption logic within reasonable computational timeframes. Even implementations such as Bitcoin and other blockchain technologies all rely on this infeasibility. However, if quantum computing were to become reality, these systems may be broken, requiring us to develop alternate methods. d